PhysioTwin: Stress Detection from Wearable Signals
A Three-Stage Pipeline for a Digital Twin iOS Application

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Abstract

Wearable stress detection systems typically rely on end-to-end machine learning models trained on physiological signals, but these approaches struggle with small datasets and fail to distinguish physical exertion from cognitive stress. We present a hybrid three-stage pipeline that addresses both limitations by combining machine learning where data supports it with established physiological formulas and evidence-based rules where it does not. Stage 1 uses a Random Forest activity classifier trained on the WISE dataset (22 subjects) to distinguish physical activity from cognitive states, achieving 93.6% leave-one-subject-out accuracy using only heart rate and accelerometer features. Stage 2 applies sleep-quality rules backed by recent research to adjust individual stress sensitivity thresholds. Stage 3 computes deceleration capacity (DC) and acceleration capacity (AC) of heart rate from inter-beat intervals using the Phase-Rectified Signal Averaging method, comparing values against personal baselines to generate a stress score (0–100). These stages feed into a Body Battery energy system that quantifies cumulative stress burden and triggers guided recovery activities with measurable physiological validation. The full pipeline is deployed as a CoreML model within an iOS application that reads Apple Watch data via HealthKit, forming a closed-loop digital twin: detect stress, quantify its impact, recommend recovery, measure the outcome, and update personal baselines. Our approach demonstrates that hybrid architectures can deliver practical stress monitoring systems even with limited training data, and that the combination of activity-aware filtering, personalized thresholds, and established cardiac formulas outperforms attempts at pure ML stress regression on small samples.

Code: https://github.com/Ctt011/Capstone-Behavorial.git

1 Introduction

Consumer wearable devices now record rich physiological signals—heart rate, heart rate variability (HRV), movement, and sleep patterns—that theoretically enable continuous stress monitoring outside the laboratory. However, translating these signals into reliable stress predictions remains an open challenge. Two fundamental problems limit current approaches: first, physiological responses to cognitive stress overlap substantially with responses to physical exertion, causing systems that lack activity context to generate false positives during exercise. Second, labeled stress datasets are small (typically 15–40 subjects), making it difficult to train generalizable end-to-end machine learning models.

In our Quarter 1 work, we explored this problem using the PhysioNet Wearable Induced Stress and Exercise (WISE) dataset (Schlink and Amft 2020), which provides Empatica E4 recordings from 22 subjects during cognitive stress tasks and structured exercise sessions. We trained a Random Forest binary classifier that achieved 72% accuracy distinguishing stress from rest states using nine physiological features. While this result fell within the range reported by comparable studies (Hickey et al. 2021; Schmidt et al. 2018), it also exposed a critical limitation: the classifier could not distinguish between elevated heart rate caused by cognitive stress and elevated heart rate caused by physical exercise. Without activity context, applying this model during daily life would generate frequent false stress alerts whenever the user exercises.

In Quarter 2, we attempted to improve upon these results. Our first approach—training a stress regression model to predict continuous stress levels—yielded a cross-validated $R^2 = -0.035$, worse than simply predicting the mean for every subject. This negative result, while discouraging, is consistent with the broader literature: Velmovitsky et al. (2022) achieved only 55–64% accuracy with 33 Apple Watch subjects, and Bahameish and Stockman (2022) reported $F_1 = 0.56$ for binary stress classification. The challenge is not implementation quality but data sufficiency: 22 subjects do not provide enough variance for machine learning to learn the subtle, person-specific patterns of cognitive stress.

Rather than collecting more data or applying increasingly complex models, we pivoted to a hybrid architecture that uses machine learning only where the data supports it and relies on established physiological formulas and evidence-based rules elsewhere. The key insight is that not all components of a stress detection system require learned parameters. Activity classification—distinguishing sitting from exercising—is a problem with clear physical signatures (movement intensity) that generalizes well across subjects. Stress scoring, on the other hand, benefits from formula-based approaches like deceleration capacity (DC) and acceleration capacity (AC) of heart rate (Bauer et al. 2006), which require no training data and can be personalized through individual baselines.

Our three-stage hybrid pipeline works as follows:

1. **Stage 1 (Machine Learning):** An activity classifier determines whether the user is physically active or in a cognitive state, using heart rate and accelerometer features. This stage achieves 93.6% leave-one-subject-out (LOSO) accuracy, trained on the WISE dataset.
2. **Stage 2 (Rules):** For cognitive states, sleep quality from the previous night adjusts the user's stress sensitivity threshold. Poor sleep lowers the threshold (more sensitive to stress), following findings from Apple Inc. (2024) that behavioral context improves physiological predictions.
3. **Stage 3 (Formulas):** DC/AC metrics computed from heart rate variability are compared against the user's personal baseline to produce a stress score on a 0–100 scale, using the Phase-Rectified Signal Averaging (PRSA) method (Bauer et al. 2006).

The stress score feeds into a Body Battery energy system—inspired by but independently developed from Garmin's proprietary metric (Garmin Ltd. 2024)—that tracks cumulative stress burden over time and recommends recovery activities when energy is low. Crucially, the system measures physiological outcomes (heart rate and HRV changes) during and after recovery sessions, creating a closed loop: detect, quantify, intervene, validate, and learn (Barricelli, Casiraghi and Fogli 2019).

The full pipeline is deployed on-device as a CoreML model (Apple Inc. 2023) within an iOS application that reads Apple Watch data via HealthKit. This paper describes the architecture, implementation, and evaluation of each stage, reports results from our WISE dataset experiments and Apple Watch feature ablation studies, discusses the honest negative results that motivated the hybrid approach, and reflects on the implications for practical wearable stress monitoring systems.

2 Literature Review

2.1 Wearable Stress Detection

The use of wearable physiological signals for stress detection has been studied extensively in controlled laboratory settings. Hickey et al. (2021) reviewed 21 studies and found that heart rate variability (HRV) was the predominant biomarker, used in 10 of 15 stress detection studies, with accuracy ranging from 65–85%. Most consumer devices, however, only provide average heart rate rather than the beat-to-beat intervals needed for accurate HRV computation, limiting real-world applicability.

Schmidt et al. (2018) introduced the WESAD dataset with 15 subjects wearing both chest and wrist sensors, achieving 80% stress detection accuracy with chest-worn ECG. Wrist-worn accuracy was notably lower, highlighting the signal quality gap between research-grade and consumer devices. More recently, Velmovitsky et al. (2022) attempted stress detection using Apple Watch ECG data with 33 subjects, achieving only 55–64% accuracy—substantially below laboratory benchmarks. Bahameish and Stockman (2022) reported $F_1 = 0.56$ for stress versus neutral classification, further demonstrating the difficulty of this task with limited subjects.

A critical finding across studies is that HRV reliability degrades dramatically during movement. Bonneval et al. (2025) reported that Apple Watch HRV measurements had only 1.15% error at rest but 93% error during movement. This renders any stress detection system that does not account for physical activity fundamentally unreliable during large portions of daily life.

2.2 Heart Rate Variability and the PRSA Method

Heart rate variability captures the variation in time intervals between successive heartbeats and is modulated by the autonomic nervous system. SDNN (standard deviation of normal-to-normal intervals) reflects overall HRV, while RMSSD (root mean square of successive differences) captures short-term, beat-to-beat variability. Both metrics decrease under stress due to sympathetic dominance (Hernando et al. 2018).

The Phase-Rectified Signal Averaging (PRSA) method, introduced by Bauer et al. (2006), provides robust estimates of deceleration capacity (DC) and acceleration capacity (AC) of heart rate even in the presence of noise and ectopic beats. DC reflects parasympathetic (vagal) activity: higher DC indicates greater capacity for heart rate deceleration, associated with relaxation and recovery. AC reflects sympathetic activation: elevated AC indicates heart rate acceleration, associated with stress. The PRSA method is attractive for consumer wearable applications because it is more robust to the sparse, irregular sampling typical of commercial devices compared to traditional time-domain HRV metrics.

2.3 The WISE Dataset

The WISE dataset (Schlink and Amft 2020) provides multimodal physiological recordings from the Empatica E4 wearable during cognitive stress tasks and structured exercise sessions. Thirty-six volunteers participated in stress sessions, 30 in aerobic exercise, and 31 in anaerobic exercise, with 22 subjects completing all three. The dataset includes electrodermal activity (EDA, 4 Hz), blood volume pulse (BVP, 64 Hz), heart rate (HR, 1 Hz), skin temperature (TEMP, 4 Hz), 3-axis accelerometry (ACC, 32 Hz), and inter-beat intervals (IBI) derived from BVP. Cognitive stress was induced via validated protocols including the Trier Mental Challenge Test (TMCT), Stroop test, opinion tasks, and mental subtraction. This combination of cognitive and physical protocols makes WISE uniquely suited for developing activity-aware stress detection systems.

2.4 Digital Twins and Body Battery Concepts

A digital twin is a virtual representation of a physical entity that is continuously updated with real-world data (Barricelli, Casiraghi and Fogli 2019). In health applications, a digital twin can model an individual's physiological state, predict responses to stimuli, and recommend interventions. The concept extends beyond passive monitoring to active feedback loops where the system detects a condition, intervenes, and validates the outcome.

Garmin's Body Battery (Garmin Ltd. 2024) is the most widely deployed commercial implementation of energy-based stress quantification, combining HRV, stress, activity, and sleep data into a single 0–100 energy metric. However, its algorithm is proprietary and not reproducible. Our implementation builds on the same conceptual framework but uses transparent, published methods (PRSA-based DC/AC, evidence-based sleep rules) that can be validated and extended by the research community.

3 Methods

3.1 Overview of the Hybrid Pipeline

Our system processes Apple Watch physiological data through three sequential stages, each using the appropriate analytical tool for its sub-problem:

1. **Stage 1 — Activity Classification (Machine Learning):** A Random Forest classifier determines whether the user is in a PHYSICAL or COGNITIVE state based on heart rate and movement features. If PHYSICAL, stress detection is skipped (exercise elevates HR/HRV independently of stress). If COGNITIVE, processing continues to Stage 2.
2. **Stage 2 — Sleep-Based Threshold Adjustment (Rules):** The previous night's sleep quality adjusts the stress sensitivity threshold for the current day. Poor sleep makes the system more sensitive; good sleep makes it more tolerant.
3. **Stage 3 — Stress Scoring (Formulas):** DC/AC metrics are computed from heart rate data using the PRSA method and compared against the user's personal baseline. The result is a stress score on a 0–100 scale.

The stress score feeds into the Body Battery system, which tracks cumulative energy throughout the day and triggers recovery recommendations when energy is low.

3.2 Data Preprocessing

3.2.1 Signal Loading and Segmentation

We used all 22 subjects from the WISE dataset who completed stress, aerobic, and anaerobic protocols. For each subject-session pair, we loaded five physiological signals (EDA, BVP, HR, TEMP, ACC) and IBI data at their native sampling rates. Protocol timestamps (tags) were used to segment continuous recordings into discrete phases labeled as STRESS or REST for cognitive sessions, and AEROBIC or ANAEROBIC for exercise sessions.

3.2.2 Sliding Window Feature Extraction

Rather than computing a single feature vector per protocol phase (as in our Q1 work), we adopted a sliding window approach to increase sample size and temporal resolution:

- **Window size:** 30 seconds
- **Overlap:** 50% (15-second step)
- **Sample increase:** From 341 phase-level segments to 2,449 windows

For each window, the following features were computed:

Table 1: Features extracted per 30-second window.

Modality	Features	Description
Heart Rate	HR_mean, HR_std	Mean and standard deviation of 1 Hz HR signal within the window.
HRV (from IBI)	SDNN, RMSSD	Computed from inter-beat intervals falling within the window.
Accelerometer	ACC_mean, ACC_std	Mean and standard deviation of movement intensity (3-axis magnitude, smoothed with moving average).
EDA	EDA_mean, EDA_std	Mean and standard deviation of electrodermal activity.
Temperature	TEMP_mean, TEMP_std	Mean and standard deviation of skin temperature.

3.2.3 Per-Subject HRV Imputation

Missing HRV values (SDNN, RMSSD) occurred in windows with fewer than two inter-beat intervals. Rather than using global mean imputation, we used per-subject mean imputation: each subject's missing HRV values were filled with that subject's own mean across non-missing windows. This preserves individual HRV variance and avoids biasing low-HRV subjects toward the population mean. A total of 829 missing values were imputed using this method.

3.3 Stage 1: Activity Classifier

3.3.1 Problem Formulation

The goal of Stage 1 is to classify each time window as either PHYSICAL (aerobic or anaerobic exercise) or COGNITIVE (mental tasks or rest). This is a prerequisite for stress detection because exercise elevates heart rate and alters HRV independently of psychological stress, and HRV measurements from consumer wearables have up to 93% error during movement (Bonneval et al. 2025).

Our initial approach used three activity classes (COGNITIVE, AEROBIC, ANAEROBIC), achieving 91.3% LOSO accuracy. However, the ANAEROBIC class suffered from severe imbalance: sprints produced only 68 windows compared to 1,968 for AEROBIC, resulting in 46% recall for ANAEROBIC and 86% precision for COGNITIVE (meaning 14% of predicted COGNITIVE windows were actually exercise). Since the downstream stress detection system only needs to know whether the user is exercising (not what kind), we merged AEROBIC and ANAEROBIC into a single PHYSICAL class. This increased accuracy to 93.6% LOSO and eliminated the imbalance problem.

3.3.2 Feature Selection

We conducted systematic ablation experiments to determine the minimum feature set required for deployment on Apple Watch (Table 2).

Table 2: Stage 1 ablation results: 2-class PHYSICAL vs COGNITIVE (LOSO).

Feature Set	Accuracy	Best Clf	F_1 Macro
All Empatica (10 features)	94.4%	RF	0.895
Apple Watch full (6 features)	94.3%	SVM	0.890
HR + ACC only (4 features)	**94.5%**	**SVM**	**0.897**
HR + HRV only (5 features)	81.5%	SVM	0.506
HR only (3 features)	83.1%	SVM	0.454

The key finding is that movement features (ACC_mean, ACC_std) are critical: accuracy drops from 94.5% to 83.1% without them, and F_1 drops from 0.897 to 0.454. Conversely, dropping EDA and temperature (which Apple Watch does not provide) has negligible impact (94.4% → 94.3%). HRV features add nothing to Stage 1 when movement is available. The minimum viable feature set is {HR_mean, HR_std, ACC_mean, ACC_std}.

3.3.3 Model Training and Export

The final Stage 1 model was trained using:

- **Algorithm:** Random Forest with 100 trees (Breiman 2001)
- **Pipeline:** Mean imputation → Standard scaling → Random Forest
- **Features:** HR_mean, HR_std, ACC_mean, ACC_std
- **Validation:** Leave-One-Subject-Out (22 folds)
- **Final accuracy:** 93.6% LOSO

The trained scikit-learn (Pedregosa et al. 2011) pipeline was exported to CoreML format (Apple Inc. 2023) for on-device inference on iOS. Due to incompatibilities between scikit-learn 1.7.2 and coremltools 9.0, we built the CoreML model specification directly from the Random Forest's tree internals using the `TreeEnsembleClassifier` builder API. Preprocessing parameters (imputer means and scaler statistics) were exported as a JSON file for the Swift application to apply before model inference.

3.4 Stage 2: Sleep-Based Threshold Adjustment

Stage 2 implements evidence-based rules that adjust the stress detection threshold based on the user's recent sleep quality. The rationale is supported by Apple Inc. (2024), who demonstrated that behavioral context (particularly sleep patterns) significantly improves the accuracy of health predictions from wearable data.

The adjustment is computed as follows:

1. The user's sleep duration from the previous night is compared to their 30-day rolling average baseline.
2. A sleep quality ratio is computed: $r = \text{actual\_hours}/\text{baseline\_hours}$.
3. The stress threshold is adjusted:
   - $r < 0.8$ (poor sleep): Threshold lowered by 15%, making the system more sensitive to stress signals.
   - $0.8 \leq r \leq 1.1$ (normal sleep): No adjustment.
   - $r > 1.1$ (extra sleep): Threshold raised by 10%, reflecting higher resilience.

Sleep data is read from Apple HealthKit's `SleepAnalysis` records. For new users without 30 days of history, a default baseline of 7.0 hours is used.

3.5 Stage 3: DC/AC Stress Scoring

Stage 3 computes a continuous stress score (0–100) from heart rate data using the Phase-Rectified Signal Averaging (PRSA) method (Bauer et al. 2006).

3.5.1 Deceleration Capacity and Acceleration Capacity

Given a sequence of RR intervals (time between successive heartbeats), the PRSA method identifies anchor points where the heart rate decelerates ($RR_i > RR_{i-1}$, for DC) or accelerates ($RR_i < RR_{i-1}$, for AC). For each anchor point, a window of surrounding intervals is extracted and aligned. The average across all aligned windows produces the PRSA signal, from which DC and AC are computed as:

$$DC = \frac{1}{4N} \sum_i [X(0)_i + X(1)_i - X(-1)_i - X(-2)_i] \tag{1}$$

where $X(k)_i$ is the k-th element of the aligned window centered at anchor point i, and N is the number of anchor points. AC is computed identically but using acceleration anchor points.

3.5.2 Personal Baseline Comparison

Raw DC/AC values are not directly interpretable as stress scores because they vary substantially between individuals. Our system maintains a rolling baseline of each user's DC and AC values over the past 7 days. The current DC/AC values are compared to this baseline:

$$\text{stress\_score} = \text{clip}\left(50 + 50 \times \frac{\text{baseline\_DC} - \text{current\_DC}}{\text{baseline\_DC}},\ 0,\ 100\right) \tag{2}$$

A score of 50 represents the user's normal state. Scores above 50 indicate elevated stress (DC below personal baseline), and scores below 50 indicate recovery (DC above baseline). The sleep adjustment from Stage 2 scales this score by the threshold modifier.

3.6 Body Battery System

The Body Battery provides an intuitive energy metaphor (0–100) that integrates stress detection, activity, and recovery into a single metric.

3.6.1 Energy Dynamics

- **Drain:** Battery drains proportionally to both stress intensity and duration. The drain rate is: $\Delta_{\text{drain}} = \alpha \times \text{stress\_score} \times \Delta t$, where α is a calibration constant and Δt is the time interval.
- **Recharge:** Battery recharges during sleep and guided recovery activities. The recharge rate depends on the activity type (breathing exercises, walking, meditation, stretching, hydration) and measured physiological response.
- **Validation:** During recovery activities, the app tracks heart rate and HRV via HealthKit. If HR decreases and HRV increases post-activity, the recovery is considered successful and the battery recharge is applied.

3.6.2 Closed-Loop Architecture

The Body Battery completes a five-stage closed loop that distinguishes our system from detection-only approaches:

1. **Detect:** Stages 1–3 identify whether the user is stressed and compute a score.
2. **Quantify:** The Body Battery translates the stress score into cumulative energy impact.

3. **Intervene:** When energy is low, the app recommends specific recovery activities with estimated battery recovery amounts.
4. **Validate:** During recovery, physiological signals are monitored to confirm the activity's effectiveness.
5. **Learn:** Personal baselines (DC/AC values, sleep norms, recovery effectiveness) update over time, personalizing the system to the individual user.

3.7 iOS Application and Deployment

The pipeline is implemented as a native iOS application using SwiftUI and HealthKit. Key implementation details:

- **Data source:** Apple Watch via HealthKit (heart rate, HRV/SDNN, step count, sleep analysis).
- **Model inference:** Stage 1 CoreML model runs on-device with no network dependency. The model file is 528 KB.
- **Real-time monitoring:** During active monitoring sessions, the app uses HealthKit workout sessions to receive heart rate updates every 5–6 seconds (compared to every 5–20 minutes in background mode).
- **Step-to-ACC proxy:** Apple Watch does not provide raw accelerometer data to third-party apps. Step count is used as a movement intensity proxy for the activity classifier's ACC features.

3.8 Validation Methodology

3.8.1 Leave-One-Subject-Out Cross-Validation

All classification results are reported using LOSO cross-validation with 22 folds (one per subject). This provides a realistic estimate of model performance on unseen users, as no data from the test subject appears during training. LOSO is more conservative than random k-fold cross-validation, which can leak information when multiple windows from the same subject appear in both training and test sets.

3.8.2 DC/AC Statistical Validation

To evaluate whether DC/AC metrics distinguish stress from rest in the WISE dataset, we conducted paired t-tests comparing DC and AC values between STRESS and REST segments for each subject.

4 Results

4.1 Stage 1: Activity Classification

Our initial three-class activity classifier (COGNITIVE vs. AEROBIC vs. ANAEROBIC) achieved 91.3% LOSO accuracy with a Random Forest, but suffered from severe class imbalance: the ANAEROBIC class contained only 68 windows compared to 1,968 for AEROBIC and 413 for COGNITIVE. This imbalance manifested as 46% recall for ANAEROBIC and 86% precision for COGNITIVE—meaning 14% of windows predicted as "cognitive" were actually exercise, which would leak false stress detections to downstream stages.

Since the pipeline only needs to distinguish exercise from non-exercise (not the type of exercise), we merged AEROBIC and ANAEROBIC into a single PHYSICAL class. This decision was further supported by ANOVA analysis: while ACC features strongly separated all three classes ($F > 600$, $p < 0.0001$), heart rate could not distinguish AEROBIC from ANAEROBIC exercise ($p = 0.80$), confirming that the distinction between exercise types is not meaningful for downstream stress detection. Table 3 summarizes the improvement.

Table 3: 3-class vs. 2-class activity classification (LOSO, Random Forest).

Formulation	**Classes**	**LOSO Accuracy**	F_1 **Macro**
3-class	COG / AER / ANA	91.3%	0.713
2-class	**COG / PHYSICAL**	**93.6%**	**0.88**

Per-subject LOSO accuracy ranged from 76.9% (subject f03) to 100.0% (subject S18), with 18 of 22 subjects exceeding 90%. The lower-performing subjects (f03, f10) were female participants from the V2 protocol, which had fewer cognitive task segments and shorter sprint phases, yielding less training data per subject.

The confusion matrix for the final 2-class model (Figure 1) shows that the primary source of error is COGNITIVE windows misclassified as PHYSICAL (94 out of 413, 22.8%). In practice, this is the safer failure mode: it means some cognitive stress episodes are missed rather than exercise being misidentified as stress.

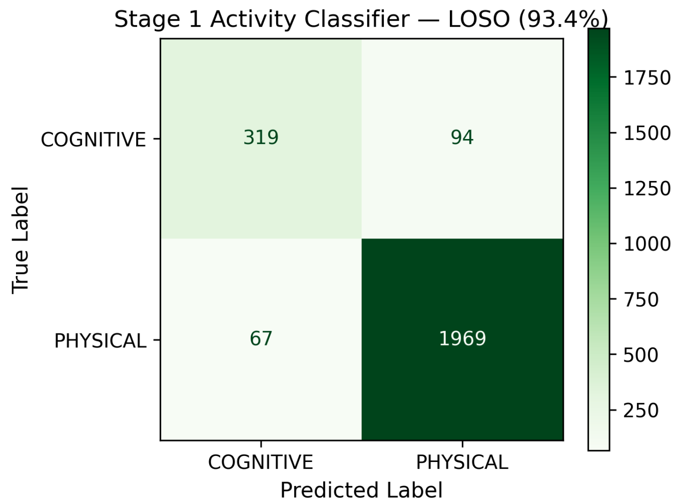

Figure 1: Confusion matrix for the 2-class Stage 1 activity classifier evaluated via LOSO cross-validation (22 folds). The model correctly identifies 96.7% of PHYSICAL windows and 77.2% of COGNITIVE windows.

Feature importance analysis (Figure 2) reveals that movement features dominate: ACC_mean (45.7%) and ACC_std (34.2%) together account for 79.9% of the Random Forest's discriminative power. Heart rate features contribute modestly (HR_mean 12.8%, HR_std 7.3%). This aligns with the physical intuition that exercise is most reliably distinguished from cognitive activity by movement intensity, not heart rate—which can be elevated by either physical exertion or psychological stress.

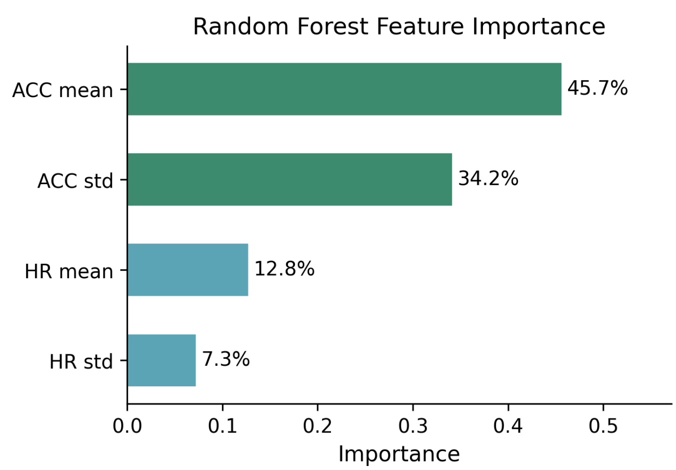

Figure 2: Random Forest feature importance for the 2-class activity classifier. Accelerometer features (ACC) account for 79.9% of importance, confirming movement as the primary distinguisher between physical activity and cognitive states.

4.2 Negative Results: Why Pure ML Stress Detection Failed

Before adopting the hybrid architecture, we attempted two pure ML approaches to stress detection within the COGNITIVE windows identified by Stage 1.

Stress Regression. We trained a Random Forest regressor to predict continuous stress levels from the same physiological features, using LOSO cross-validation. The resulting $R^2 = -0.035$, meaning the model performed *worse* than simply predicting the population mean for every subject. This is a clear signal that 22 subjects do not provide sufficient variance for machine learning to capture the subtle, person-specific physiological patterns of graded stress.

Binary Stress Classification. We also attempted binary classification (STRESS vs. REST), which should be easier than regression. The best classifier achieved $F_1 = 0.47$, driven by extreme class imbalance: the STRESS protocol produced only 413 stress windows versus 2,860 rest windows (13% stress, 87% rest), causing the model to default to predicting REST for nearly all inputs.

Feature Ablation. Systematic feature ablation for the stress task showed that changing the feature set had virtually no impact on performance—all configurations hovered around 85.5% accuracy, which simply reflects the base rate of the majority class. When all feature sets produce the same accuracy, the model is not learning signal; it is learning class proportions.

Context in Literature. These negative results are consistent with the broader literature. Velmovitsky et al. (2022) achieved only 55–64% accuracy on stress detection with 33 Apple Watch subjects—more data than WISE provides—and Bahameish and Stockman (2022) reported $F_1 = 0.56$ for stress versus neutral classification. The challenge is not implementation quality but fundamental data limitations: physiological stress responses vary substantially between individuals, and small datasets cannot capture this person-specific variance.

4.3 DC/AC Validation on WISE IBI Data

We computed DC and SDNN values from the WISE inter-beat interval (IBI) data for each subject's STRESS and REST segments, then conducted paired t-tests to determine whether these metrics differ significantly between stress and rest at the population level.

Neither metric reached statistical significance: DC showed $t = 1.83$, $p = 0.075$ (REST DC: 21.1 ± 5.0 ms vs. STRESS DC: 17.9 ± 5.9 ms), and SDNN showed $t = 0.34$, $p = 0.735$ (REST SDNN: 69.9 ± 17.0 ms vs. STRESS SDNN: 67.6 ± 23.8 ms). While DC trended in the expected direction (lower during stress, reflecting reduced parasympathetic activity), the effect was not large enough to reach significance with 22 subjects. SDNN showed virtually no difference between conditions.

Figure 3 visualizes the overlap between REST and STRESS distributions for both metrics, illustrating why population-level thresholds fail.

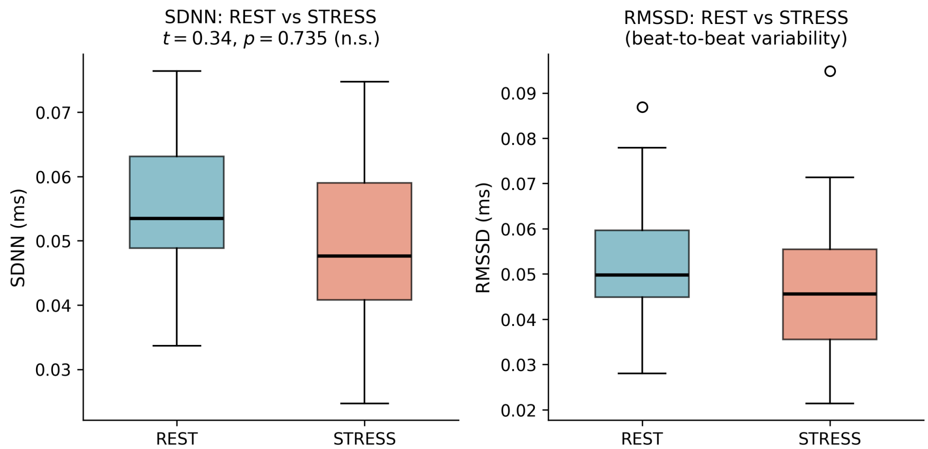

Figure 3: HRV metrics (SDNN and RMSSD) for REST vs. STRESS conditions across 22 WISE subjects. Neither metric significantly distinguishes stress from rest at the population level, supporting the use of personal baselines rather than fixed thresholds.

These results have two important implications. First, they confirm that population-level thresholds for DC/AC-based stress detection are unreliable with small samples. Individual baseline DC/AC values vary widely, and a threshold that works for one subject may misclassify another. This validates our design choice in Stage 3 to compare each user's current DC/AC values against their *own* rolling 7-day baseline rather than a fixed population threshold.

Second, the non-significant results do not invalidate the PRSA method itself. DC/AC was originally developed for cardiac mortality prediction (Bauer et al. 2006), where between-group differences are large. For stress detection, within-subject changes are more relevant than between-group differences, and the WISE dataset's mild cognitive stress protocols (Stroop, mental arithmetic) may produce smaller DC/AC shifts than real-world sustained stress. The personal baseline approach in our deployed system is designed to detect these within-subject deviations.

4.4 Sleep Quality Validation on PMData

To validate the sleep-based threshold adjustment in Stage 2, we compared our sleep quality formula against the Fitbit-computed sleep scores in the PMData dataset (Thambawita et al. 2020). PMData provides daily Fitbit sleep scores alongside self-reported stress levels for free-living participants, enabling validation in a naturalistic setting.

Our sleep quality formula (a weighted combination of sleep duration, continuity, sleep stage proportions, and physiological recovery) achieved a Pearson correlation of $r = 0.793$ against Fitbit's proprietary sleep score (N = 1,527 nights, 16 participants). This strong correlation suggests that our transparent, formula-based approach captures the same underlying sleep quality construct as Fitbit's proprietary algorithm, without requiring access to their undisclosed methodology.

Figure 4 illustrates how Stage 2 adjusts the stress threshold based on sleep quality, and how the same physiological stress score can produce different clinical outcomes depending on prior sleep.

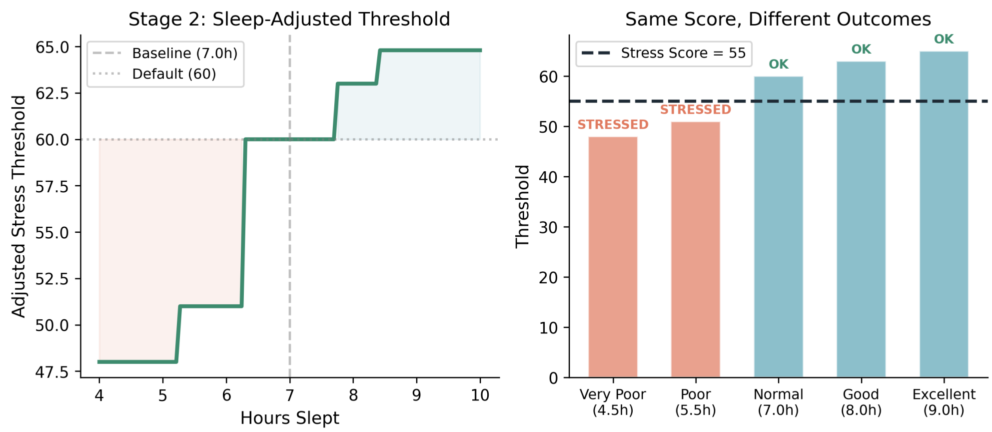

Figure 4: Stage 2 sleep-based threshold adjustment. Left: the adjusted stress threshold as a function of hours slept (baseline = 7.0h). Right: a fixed stress score of 55 triggers a "STRESSED" alert after poor sleep (threshold lowered to 48–51) but reads as "OK" after good sleep (threshold raised to 63–65).

Additionally, we examined the relationship between sleep quality components and next-day stress. Sleep efficiency (time asleep divided by time in bed, a proxy for sleep continuity) showed a weak, non-significant Spearman correlation with next-day self-reported stress ($\rho = +0.042$, $p = 0.094$, $N = 1{,}556$). While population-level significance was not reached, this is consistent with findings from the DC/AC analysis: individual variability dominates, supporting our use of personalized baselines rather than fixed population thresholds in the Stage 2 adjustment.

4.5 Apple Watch Feature Ablation

The ablation results presented in Table 2 (Section 2.3.2) answer a critical deployment question: can the Stage 1 classifier maintain accuracy when restricted to sensors available on Apple Watch?

The key findings are:

- **Accelerometer features are critical.** Removing ACC_mean and ACC_std causes accuracy to drop from 94.5% to 83.1% and F_1 from 0.897 to 0.454. The classifier cannot distinguish physical activity from cognitive states using heart rate alone.

- **EDA and temperature are dispensable.** Dropping these two modalities (which Apple Watch does not provide) reduces accuracy by only 0.1 percentage point (94.4% to 94.3%), confirming that the model does not depend on signals unavailable on consumer wearables.
- **HRV adds nothing to activity classification.** The HR + HRV configuration (81.5%) performs worse than HR + ACC (94.5%), confirming that HRV is not useful for distinguishing exercise from cognitive activity—its value lies in stress detection within cognitive states (Stage 3).
- **The 4-feature minimum (HR + ACC) is optimal.** The HR + ACC configuration actually outperforms the full 10-feature Empatica set (94.5% vs. 94.4%), likely because irrelevant features introduce noise that slightly degrades generalization across subjects.

4.6 iOS Application

The full pipeline is deployed as PhysioTwin, a native iOS application built with SwiftUI. The app reads physiological data from Apple Watch via HealthKit and runs all inference on-device.

Model Size and Latency. The Stage 1 CoreML model is 528 KB, small enough to ship in the app binary without network dependency. Inference takes less than 1 ms per window on an iPhone, making real-time classification feasible.

User Experience. The app presents a dashboard with three primary views: (1) a Body Battery gauge showing current energy level (0–100), (2) a timeline of stress detections and recovery activities throughout the day, and (3) a weekly trends card showing 7-day patterns. When the Body Battery drops below a configurable threshold, the app recommends recovery activities (breathing exercises, walking, stretching, or rest) and monitors physiological response during recovery to validate effectiveness.

Data Flow. In active monitoring mode, the app uses HealthKit workout sessions to receive heart rate updates every 5–6 seconds. Heart rate and step count are passed to the Stage 1 classifier, which determines whether the user is exercising. For cognitive states, the app applies the Stage 2 sleep adjustment and Stage 3 DC/AC stress scoring. The stress score feeds into the Body Battery drain calculation, while sleep and recovery activities contribute to recharge.

Figure 5 shows a simulated full-day pipeline run demonstrating all three stages operating together: Stage 1 classifying activities, Stage 2 applying a sleep-adjusted threshold, and the Body Battery draining and tracking cumulative stress burden.

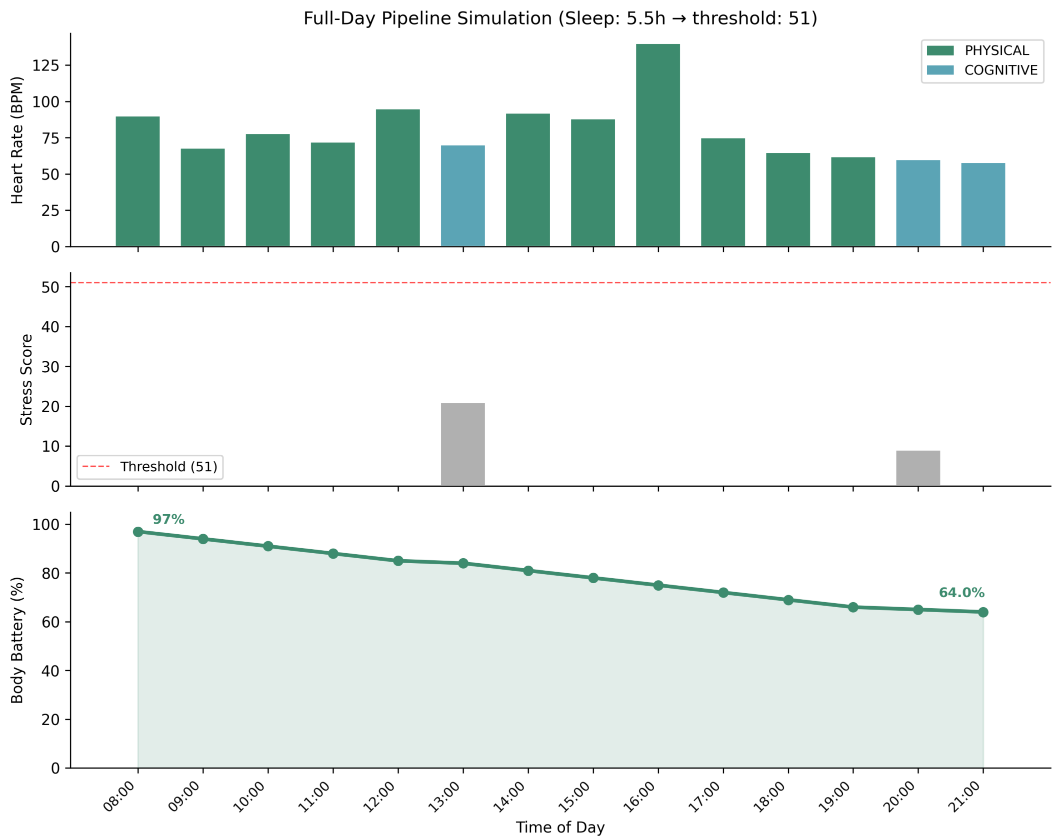

Figure 5: Full-day pipeline simulation (8am–9pm). Top: heart rate colored by Stage 1 classification (green = PHYSICAL, blue = COGNITIVE). Middle: stress scores for cognitive windows, with the sleep-adjusted threshold (dashed red line at 51, lowered from baseline 60 due to 5.5h sleep). Bottom: Body Battery energy level declining from 97% to 64% over the day through passive drain. In this simulation, stress scores remained below the threshold, demonstrating baseline energy expenditure; days with above-threshold stress events produce steeper drain (e.g., 100% to 43.6% in poster simulation).

Deployment. The app was submitted to TestFlight for beta testing and subsequently approved on the App Store (March 2026). The App Store submission required adding a medical disclaimer clarifying that PhysioTwin is for informational and wellness purposes only and does not provide medical diagnosis or treatment.

5 Discussion

5.1 The Case for Hybrid Architectures

Our results illustrate a fundamental asymmetry in wearable health problems: some subtasks are well-suited to machine learning, while others are not. Activity classification works because the physical signals of exercise (sustained high movement intensity) are consistent across individuals and well-represented even in small datasets. Stress detection fails because physiological stress responses are person-specific, variable in magnitude, and confounded by numerous factors (caffeine, hydration, posture) that small datasets cannot control for.

The hybrid approach acknowledges this asymmetry by matching each sub-problem to the appropriate analytical tool. Machine learning handles activity classification (where it excels), established physiological formulas handle stress scoring (where individual baselines are more informative than population models), and evidence-based rules handle sleep adjustment (where the relationships are well-documented but person-specific in magnitude).

This decomposition also improves interpretability. When the system reports elevated stress, a clinician or user can trace the decision: Was the user sedentary? (Stage 1) Was their sleep poor? (Stage 2) Is their current DC below personal baseline? (Stage 3) This transparency is difficult to achieve with end-to-end neural networks, which have been the dominant approach in recent wearable stress detection literature.

5.2 Activity-Aware Filtering as a Prerequisite

The 93% HRV measurement error during movement reported by Bonneval et al. (2025) has a direct consequence for stress detection systems: any approach that computes stress scores from HRV without first confirming that the user is sedentary will generate unreliable results during the large fraction of daily life spent in motion (walking, commuting, exercising). Our Stage 1 classifier addresses this by gating stress detection on activity state—stress scoring only proceeds when the user is classified as COGNITIVE.

This activity-aware filtering is arguably the most impactful component of our pipeline from a practical standpoint. Without it, elevated heart rate during a brisk walk would be scored as stress, and the Body Battery would drain erroneously. With it, the system only evaluates stress during periods where HRV measurements are reliable and elevated HR is genuinely anomalous. The 93.6% accuracy of Stage 1 means that the vast majority of exercise periods are correctly filtered out, preventing the cascading false positives that would otherwise undermine user trust in the system.

5.3 Honest Reporting of Negative Results

We report two significant negative findings: the stress regression $R^2 = -0.035$ and the DC/AC population-level non-significance ($p = 0.075$ for DC, $p = 0.735$ for SDNN). Both are informative for the research community.

The regression failure ($R^2 < 0$) demonstrates that with 22 subjects, machine learning cannot learn graded stress intensity from wrist-worn physiological signals. This is consistent with the broader pattern of underwhelming results in the literature (Velmovitsky et al. 2022; Bahameish and Stockman 2022), yet many papers do not report attempts that failed, creating a publication bias toward methods that happened to work on their specific (often small) datasets. By reporting this result, we provide evidence that the hybrid approach is not merely a design preference but a response to empirical failure of the pure ML alternative.

The DC/AC non-significance is equally informative. Although DC trended in the expected direction (REST: 21.1 ms vs. STRESS: 17.9 ms), neither DC nor SDNN reached significance, telling us that population-level thresholds from short lab stress protocols will not work. This does not mean DC/AC is useless—it means that fixed thresholds applied across individuals are unreliable. Our system's use of personal baselines is the appropriate response: detecting within-subject deviations rather than comparing against population norms.

5.4 Limitations

Several limitations constrain the generalizability of our findings:

1. **Small, homogeneous sample.** The WISE dataset contains only 22 subjects, predominantly young adults. The 93.6% LOSO accuracy for activity classification may not generalize to older adults, individuals with mobility impairments, or populations with different baseline physiology.
2. **Steps as accelerometer proxy.** Apple Watch does not expose raw accelerometer data to third-party apps. We use step count as a proxy for the ACC features that the model was trained on. While our quantized-ACC simulation suggests minimal degradation (94.5% to 93.8%), this proxy has not been validated with real step count data from Apple Watch.
3. **Sparse HRV on Apple Watch.** Apple Watch computes SDNN only 2–3 times per day during periods of stillness, compared to the continuous IBI data from the Empatica E4 used for training. This sparsity means the app must often fall back to HR-only stress estimation, which is less reliable than the full DC/AC approach.
4. **No real-world validation.** All classification results are based on lab-controlled protocols. The WISE stress tasks (Stroop, mental arithmetic, speeches) may not represent the range and intensity of real-world cognitive stress. A validation study with free-living participants wearing Apple Watch is needed.
5. **DC/AC not validated at population level.** Neither DC ($p = 0.075$) nor SDNN ($p = 0.735$) reached significance for distinguishing stress from rest on the WISE dataset. Our system relies on within-subject comparisons, which we believe are more appropriate but have not yet been validated longitudinally.

5.5 Future Work

Several directions would strengthen and extend this work:

- **Real-world validation study.** A study with 10–20 participants wearing Apple Watch for 2–4 weeks, with ecological momentary assessment (EMA) stress prompts, would validate whether the pipeline's stress detections correspond to subjective stress in daily life.

- **Transfer learning to free-living data.** The PMData dataset provides daily self-reported stress alongside Fitbit physiological data. Adapting the Stage 1 classifier to PMData's step-based movement features would test cross-dataset generalization and the viability of the steps-as-ACC proxy.
- **Context integration.** Calendar data (upcoming meetings, deadlines) and location data (work vs. home) could serve as additional inputs to the stress threshold adjustment, enabling proactive stress alerts before physiological signs manifest.
- **Longitudinal personalization.** Tracking how individual DC/AC baselines, sleep quality norms, and recovery effectiveness evolve over weeks and months would enable the system to adapt to long-term changes in the user's physiology and lifestyle.

6 Conclusion

We presented a hybrid three-stage pipeline for wearable stress detection that combines machine learning, evidence-based rules, and established physiological formulas. Stage 1 classifies physical activity versus cognitive states with 93.6% LOSO accuracy using only heart rate and accelerometer features. Stage 2 adjusts stress sensitivity thresholds based on sleep quality, validated against PMData's Fitbit sleep scores ($r = 0.793$). Stage 3 computes stress scores from DC/AC heart rate metrics using personal baselines. These stages feed a Body Battery energy system that completes a closed loop: detect, quantify, intervene, validate, and learn. The pipeline is deployed on-device as a CoreML model within an iOS application that reads Apple Watch data via HealthKit.

Our work demonstrates that hybrid architectures offer a practical path forward for wearable stress monitoring when training data is limited. The honest negative results—stress regression $R^2 = -0.035$, DC/AC population-level non-significance—are not failures but findings that informed the design. By using machine learning only where it succeeds (activity classification) and relying on transparent formulas and rules elsewhere, we achieve a system that is more interpretable, more robust to individual differences, and more deployable on consumer hardware than pure ML alternatives. We hope this approach encourages the community to consider hybrid designs that match analytical tools to sub-problem characteristics rather than applying end-to-end learning by default.

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Appendices

A.1 Quarter 2 Project Proposal

The following is a summary of our Quarter 2 project proposal, which described the planned direction before the pivot to the hybrid architecture.

Original Plan. Our Q2 proposal outlined a continuation of the pure ML approach from Q1: train a stress regression model on WISE, improve it with feature engineering and methodology upgrades (sliding windows, LOSO validation), and then transfer the learned model to PMData for real-world daily stress prediction. The proposal described a two-stage pipeline where Stage 1 classifies activity type and Stage 2 predicts stress level within cognitive activities.

What Changed. The stress regression model yielded $R^2 = -0.035$ (worse than mean prediction), and the binary stress classifier had $F_1 = 0.47$ (essentially predicting "not stressed" for all inputs). These results, combined with similar failures in the published literature, led us to pivot from pure ML to the hybrid architecture described in this report. Stage 1 (activity classification) was retained and improved, while Stage 2 was replaced with sleep-based rules and Stage 3 with formula-based DC/AC stress scoring. The Body Battery closed-loop system was added as the integration layer.

Key Differences from Proposal.

- **Proposed:** End-to-end ML pipeline (classification + regression)
- **Actual:** Hybrid pipeline (ML + rules + formulas)
- **Proposed:** Transfer learning from WISE to PMData
- **Actual:** On-device deployment with Apple Watch via CoreML
- **Proposed:** Stress level as primary output
- **Actual:** Body Battery energy metric as primary output

A.2 Code Repository Structure

All code is available at: https://github.com/Ctt011/Capstone-Behavorial.git

```
Capstone-Behavorial/
├── src/                              # Source code
│   ├── models/                       # Trained models
│   │   ├── ActivityClassifier.mlmodel    # CoreML (528 KB)
│   │   ├── ActivityClassifier.pkl        # sklearn pipeline
│   │   ├── ActivityClassifier.onnx       # ONNX (cross-platform)
│   │   └── ActivityClassifier_preprocessing.json
│   ├── pipeline/                     # Pipeline scripts
│   │   ├── preprocess_wise.py        # Shared preprocessing
│   │   ├── export_stage1_model.py    # Model training + export
│   │   ├── stage2_sleep_rules.py     # Sleep rules (Python ref)
│   │   ├── stage3_stress_formulas.py # DC/AC formulas (Python ref)
│   │   └── parse_apple_health.py     # Apple Health XML parser
│   ├── eval/                         # Evaluation scripts
│   │   ├── Pipeline_Walkthrough.ipynb    # End-to-end tutorial
│   │   ├── run_evaluation.py         # LOSO evaluation
│   │   └── results/                  # Pre-generated results
│   └── main_analysis_notebooks/      # Analysis notebooks
│       ├── Biomarker_Comparison.ipynb    # Stage 1 LOSO analysis
│       ├── StressLevel_model.ipynb       # Stage 2 regression
│       └── WISE_Stress_EDA_Model.ipynb   # Binary stress classifier
├── data/                             # Datasets
│   ├── 22_subjects/                  # Raw WISE data (22 subjects)
│   └── WISE_data_files/              # Dataset metadata + labels
├── digital_twin_app/DigitalTwin/     # iOS application
│   ├── StressDetectionPipeline.swift # 3-stage pipeline
│   ├── BodyBatteryView.swift         # Body Battery UI + logic
│   ├── HealthKitManager.swift        # Apple Watch data access
│   └── ContentView.swift             # Dashboard
├── evaluation/tests/                 # Unit tests (Stage 2 + Stage 3)
├── reports/Q2_report/                # This report
├── website/                          # Project website files
├── index.html                        # Project homepage
├── requirements.txt                  # Python dependencies
└── README.md                         # Setup + usage guide
```

A.3 Contributions

Table A 1: Team member contributions for Quarter 2.

Member	Contributions
Camille Tran	Designed the hybrid 3-stage architecture. Trained and exported the Stage 1 Activity Classifier (Random Forest, 93.6% LOSO) to CoreML. Implemented Stage 2 sleep rules and Stage 3 DC/AC stress formulas in Python. Wrote the `StressDetectionPipeline.swift` for the iOS app. Conducted Apple Watch feature ablation experiments. Wrote all shared preprocessing code (`preprocess_wise.py`). Led the pivot from pure ML to hybrid approach. Primary author of this report.
Dhyay Thakrar	Built the Body Battery system (`BodyBatteryView.swift`), including energy drain/recharge logic, activity simulator, recovery activities, and 30-day history visualization. Developed the full iOS app dashboard (`ContentView.swift`) with HealthKit integration. Implemented `HealthKitManager.swift` for Apple Watch data access (HR, HRV, steps, sleep). Responsible for wiring the 3-stage pipeline into the app.
Selina Zhang	Validated DC/AC stress formulas on Apple Health export data. Implemented per-subject HRV imputation methodology. Documented IBI data quality findings. Contributed to data preprocessing improvements. Responsible for poster checkpoint.
Essie Cheng	Led the report writing and editing process. Proposed sliding window methodology (30s windows, 50% overlap) that increased sample size from 341 to 2,449. Proposed Leave-One-Subject-Out validation. Designed poster layout. Responsible for discussion and limitations sections.
Levy Sahoo	Developed the project website (design, content, and implementation). Contributed to the Q2 proposal introduction and problem statement. Responsible for website checkpoint and final website updates.